

Data display with ggplot2

Fundamentals of Computing and Data Display

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¹Slides draw on Grolemund, G (2018). Master the Tidyverse: An Introduction to R for Data Science.
<https://github.com/rstudio/master-the-tidyverse>

Outline

- 1 Introduction
- 2 Principles and guidelines
 - Principles of graphical integrity
 - Principles for graphical excellence
- 3 ggplot2
- 4 Resources
- 5 References

Introduction

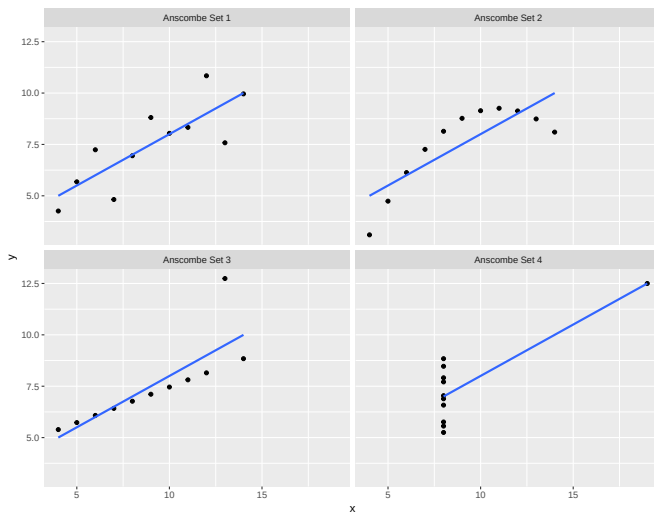
Some goals and types of data visualizations

- Info-graphics
 - Present results, monitor processes e.g. via dashboards
- Interactive exploratory analysis
 - (Web) tools for graphically exploring data
- Data analysis graphs
 - Display distributions, associations, outliers and test assumptions

→ *“Visualization tools can organize data in a meaningful way that lowers the cognitive and analytical effort required to make sense of the data and make data-driven decisions”* (Yalçın and Plaisant 2017)

Introduction

Figure: Anscombe's quartet



Principles of graphical integrity

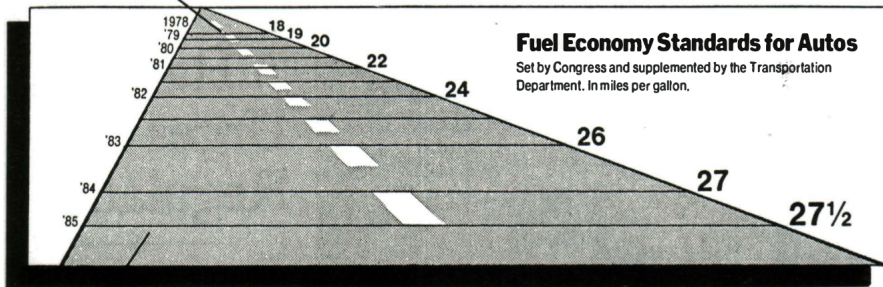
Tufte's (1983) **six principles of graphical integrity**

- ① The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the numerical quantities measured.
 - Lie Factor = $\frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$
- ② Clear, detailed, and thorough labeling should be used to defeat graphical distortion and ambiguity. Write out explanations of the data on the graphic itself. Label important events in the data.
- ③ Show data variation, not design variation
- ④ In time-series displays of money, deflated and standardized units of monetary measurement are nearly always better than nominal units.
- ⑤ The number of information-carrying (variable) dimensions depicted should not exceed the number of dimensions in the data
- ⑥ Graphics must not quote data out of context

Things to avoid

Figure: The Lie Factor

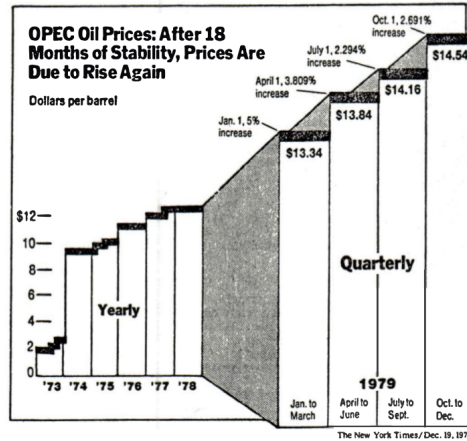
This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

Things to avoid

Figure: Design Variation



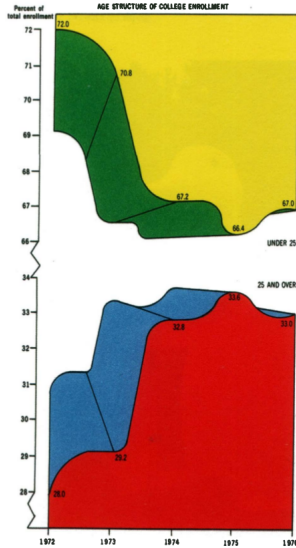
Principles for graphical excellence

Tufte's (1983) **principles for graphical excellence**

- ① Above all else show the data
- ② Maximize the data-ink ratio
 - $\text{Data ink ratio} = \frac{\text{data ink}}{\text{total ink used in the graphic}}$
- ③ Erase non-data ink
- ④ Erase redundant data ink
- ⑤ Revise and edit

Chartjunk

Figure: Self-promoting graphics – taken over by decorative forms



ggplot2

- 1 Introduction
- 2 Principles and guidelines
 - Principles of graphical integrity
 - Principles for graphical excellence
- 3 ggplot2**
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ggplot2

***“ggplot2** is a system for declaratively creating graphics, based on *The Grammar of Graphics*.”*

- Flexible: Compose a set of graphical components
- Interactive: Build plots by adding layers
- Extendable: Create new graphs based on a common theory



<https://ggplot2.tidyverse.org/>

ggplot2

Wilkinson's (2005) **Grammar of Graphics**

- “A statistical graphic is a mapping from data to aesthetic attributes (colour, shape, size) of geometric objects (points, lines, bars)” (Wickham 2016)

- ① Data: Information to be visualized
- ② Aesthetics: Description of how variables are mapped to aesthetic attributes
- ③ Geoms: (Layers of) Geometric objects that represent the data
- ④ Scales: Provide an inverse mapping of aesthetic space back to data space
- ⑤ Coordinate system: Defines the plane of the graphic
- ⑥ Facets: Describe how to split and display data subsets
- ⑦ Theme: Controls display settings

→ A layered Grammar of Graphics (Wickham 2010)

ggplot2

Key components

- **Data**

- Can be set globally when initializing the plot

- A set of **aesthetic mappings**

- Specified with mapping argument of `geom_function`

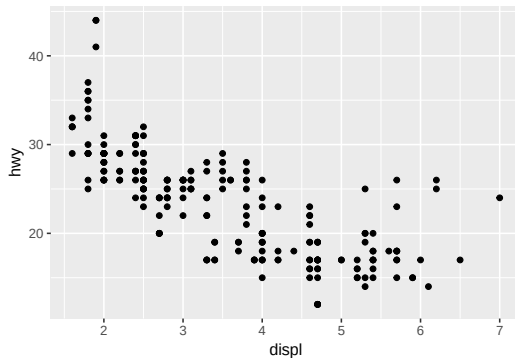
- At least one layer of **geometric objects**

- Specified with choice of `geom_function`

```
> # A code template  
> ggplot(data = <DATA>) +  
>   <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>))
```

ggplot2

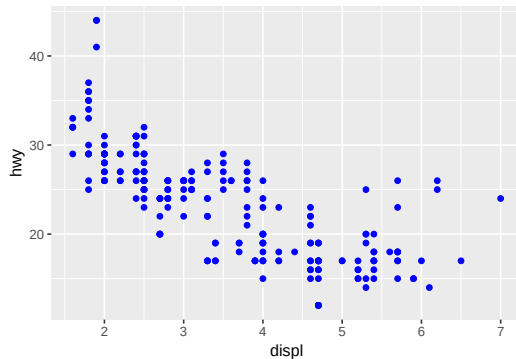
Figure: A minimalistic example



```
> ggplot(data = mpg) +  
>   geom_point(mapping = aes(x = displ, y = hwy))
```

Aesthetics

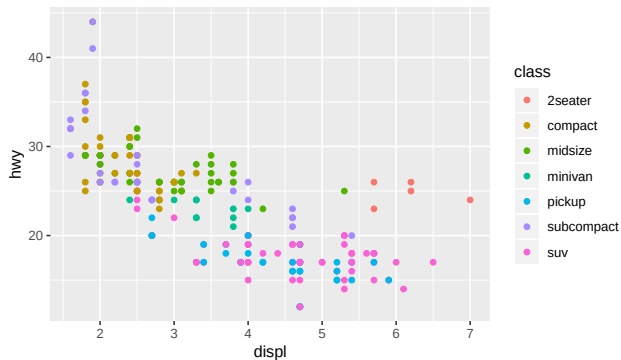
Figure: Set aesthetic to a value



```
> ggplot(data = mpg) +  
>   geom_point(mapping = aes(x = displ, y = hwy), color = "blue")
```

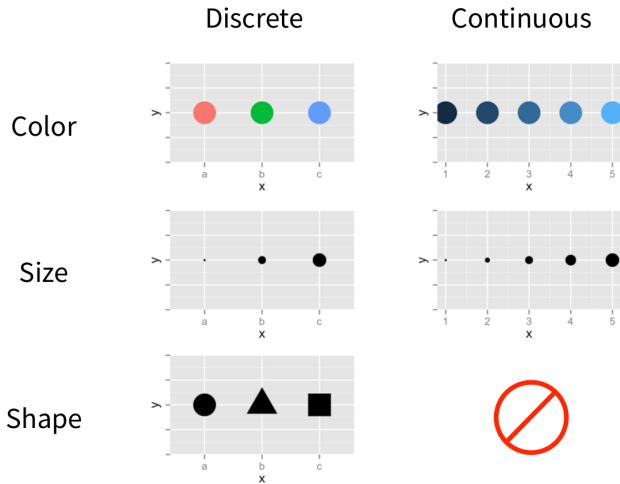
Aesthetics

Figure: Map aesthetic to a variable with color, size, shape or alpha



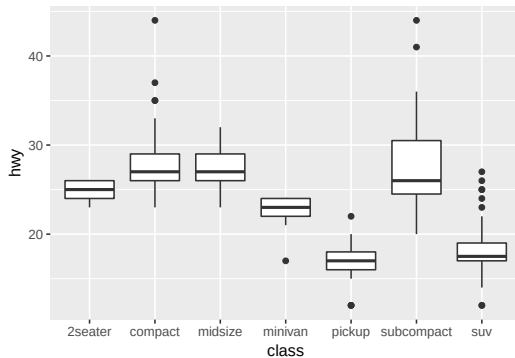
```
> ggplot(data = mpg) +  
>   geom_point(mapping = aes(x = displ, y = hwy, color = class))
```


Aesthetics



Geoms

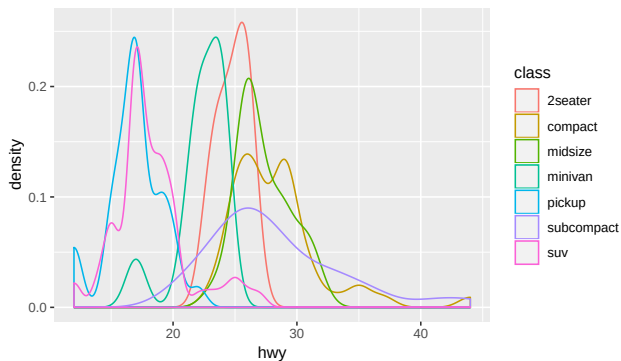
Figure: Specify geometrical representation with `geom_` functions



```
> ggplot(data = mpg) +  
>   geom_boxplot(mapping = aes(x = class, y = hwy))
```

Geoms

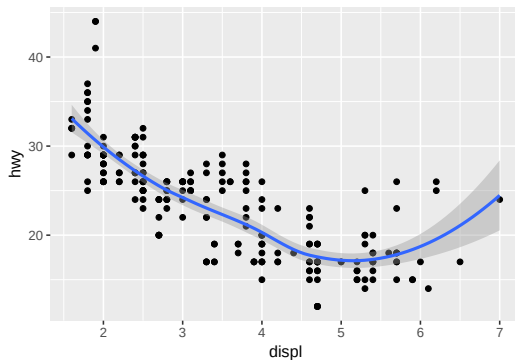
Figure: Geoms and aesthetics



```
> ggplot(data = mpg) +  
>   geom_density(mapping = aes(x = hwy, color = class))
```

Geoms

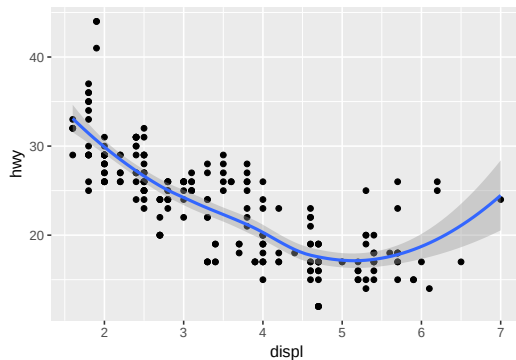
Figure: Layers of geoms



```
> ggplot(data = mpg) +  
>   geom_point(mapping = aes(x = displ, y = hwy)) +  
>   geom_smooth(mapping = aes(x = displ, y = hwy))
```

Global vs. local

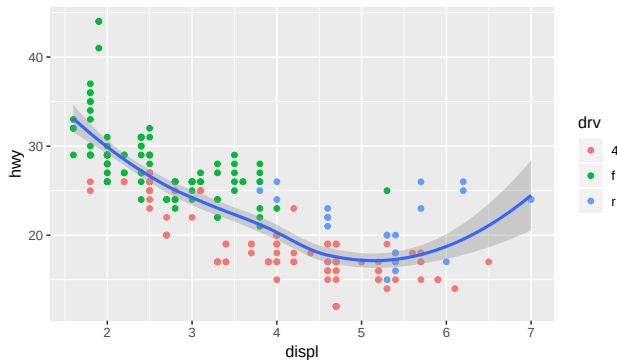
Figure: Set data and mappings globally



```
> ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +  
>   geom_point() +  
>   geom_smooth()
```

Global vs. local

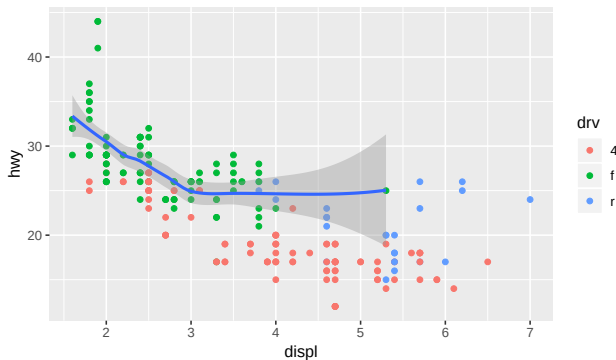
Figure: Set mappings globally and locally



```
> ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +  
>   geom_point(mapping = aes(color = drv)) +  
>   geom_smooth()
```

Global vs. local

Figure: Set data and mappings globally and locally



```
> ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +  
>   geom_point(mapping = aes(color = drv)) +  
>   geom_smooth(data = filter(mpg, drv == "f"))
```

Grammar of Graphics

Make any plot with ggplot

```
> ggplot(data = <DATA>) +  
>   <GEOM_FUNCTION> (  
>     mapping = aes(<MAPPINGS>),  
>     stat = <STAT>,  
>     position = <POSITION>  
>   ) +  
>   <COORDINATE_FUNCTION> +  
>   <FACET_FUNCTION> +  
>   <SCALE_FUNCTION> +  
>   <THEME_FUNCTION>
```


Grammar of Graphics

Figure: Build a plot from scratch I

1. Begin with the **diamonds** data set
2. Compute counts for each cut value with **stat_count()**.

carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...

stat_count()

cut	count	prop
Fair	1610	1
Good	4906	1
Very Good	12082	1
Premium	13791	1
Ideal	21551	1

Grammar of Graphics

Figure: Build a plot from scratch II

3. Represent each observation with a bar.
4. Map the **fill** of each bar to the **..count..** variable.

carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...

stat_count()

cut	count	prop
Fair	1610	1
Good	4906	1
Very Good	12082	1
Premium	13791	1
Ideal	21551	1



Grammar of Graphics

Figure: Build a plot form scratch III

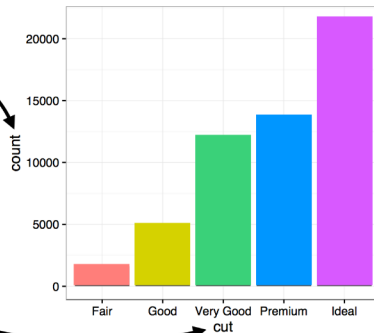
carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...

stat_count()

5. Place geoms in a cartesian coordinate system.

cut	count	prop
Fair	1610	1
Good	4906	1
Very Good	12082	1
Premium	13791	1
Ideal	21551	1

6. Map the y values to **..count..** and the x values to **cut**.



Plotting Big Data

Some solutions for big(ger) data

- Try different geoms
- Add structure with size, transparency
- Plot summary statistics, densities, model results instead of raw data
- Subsetting
- Faceting
- Sampling

Resources

- Textbook
 - Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. New York, NY: Springer.
- Galleries
 - <https://www.r-graph-gallery.com>
 - <http://r-statistics.co/Top50-Ggplot2-Visualizations-MasterList-R-Code.html>
- Cheatsheet
 - <https://github.com/rstudio/cheatsheets/raw/master/data-visualization-2.1.pdf>
- ggmap
 - <https://github.com/dkahle/ggmap>

References

- Tufte, E. R. (1983). *The Visual Display of Quantitative Information*. Cheshire, CT: Graphics Press.
- Wickham, H. (2010). A layered grammar of graphics. *Journal of Computational and Graphical Statistics* 19(1): 3–28.
- Wilkinson, L. (2005). *The grammar of graphics. Statistics and computing*. New York, NY: Springer.
- Yalçın, M. A. and Plaisant, C. (2017). Information Visualization. In Foster, I., Ghani, R., Jarmin, R. S., Kreuter, F., and Lane, J. (Eds.) *Big Data and Social Science: A Practical Guide to Methods and Tools*. Boca Raton, FL: CRC Press Taylor & Francis Group.